

Volcanoes

Few things in nature are as dramatic as volcanoes. There are many different types, ranging from simple cracks in Earth's crust to giant cone-shaped mountains built up of many layers of lava and ash. Most volcanoes form at the boundaries where tectonic plates are either coming together or pulling apart.



RIFT VOLCANOES

In the rift zones where tectonic plates pull apart, the pressure that normally keeps the mantle rock solid relaxes, allowing some of it to melt and erupt as fluid basalt lava. These eruptions are frequent and relatively predictable, but often spectacular. Erta Ale volcano in Ethiopia's Rift Valley (above) has contained a churning lake of liquid lava for more than a century.

VOLCANIC LAVA

The rivers of liquid lava erupted by rift and hot spot volcanoes can flow fast over long distances. Other volcanoes such as Mount Etna in Italy erupt stickier lava that does not flow as far. The lava erupted by volcanoes where plates are pushing together is particularly thick, and flows so slowly that it may block the volcano. Pressure then builds up to cause very explosive eruptions.

HOT SPOT VOLCANOES

Some volcanoes erupt away from plate boundaries, over hot spots below Earth's crust. At places where a plate is moving over a hot spot, new volcanoes erupt as older ones fall extinct. This creates long chains of volcanic islands such as those of Hawaii, seen here from space.



▲ ETNA
Lava erupting from Mount Etna in Italy has a temperature of about 1,830°F (1,000°C).